Questions

Q1.

Figure 5 shows the apparatus a student uses to investigate how the stopping distance of a toy car depends on the type of surface that it is stopping on.

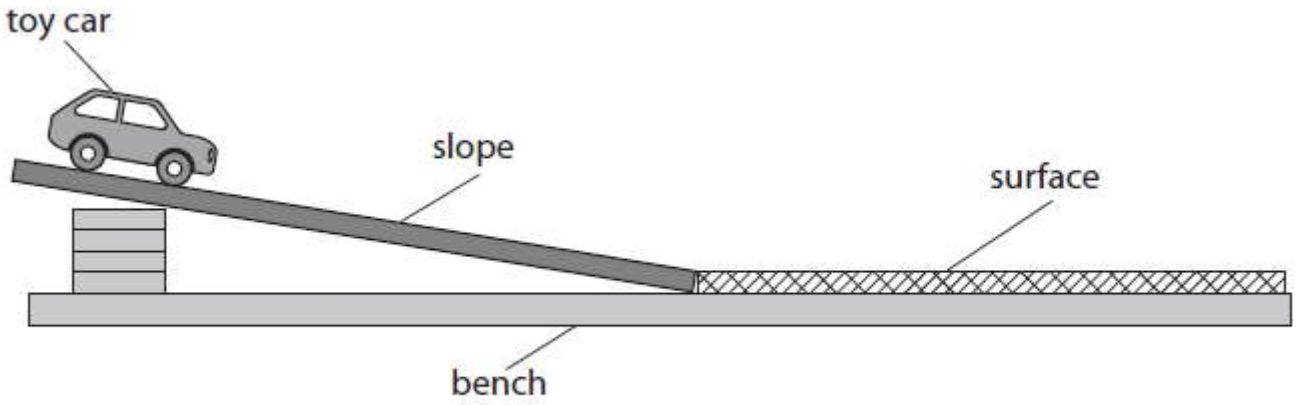

Figure 5

Figure 6 shows a set of results used to find the average stopping distance of the toy car on a surface.

test number	stopping distance in m
1	0.35
2	0.32
3	0.52
4	0.38
5	0.33

Figure 6

(i) State the anomalous value of stopping distance given in the table in Figure 6.

(1)

.....

(ii) Use the results in Figure 6 to calculate the average stopping distance.

(2)

average stopping distance = ... m

(iii) State **one** way the student could increase the speed of the car as it reaches the flat surface.

(1)

.....

.....

(Total for question = 4 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Quantities can be either scalar or vector.

Which of these is a vector quantity?

(1)

☐ **A** mass

☐ **B** force

☐ **C** energy

☐ **D** distance

(Total for question = 1 mark)

Q3.

(a) Which of these situations can increase the reaction time of a driver?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** an icy road
☐ **B** worn tyres on his car
☐ **C** stopping for a cup of coffee
☐ **D** driving for a long time without taking a break

(b) (i) A car engine produces an average driving force of 1200 N.

The car travels 8.0 m.
Calculate the work done by the force over this distance.

(2)

work done = ... J

(ii) The car has a mass of 1400 kg and travels at a velocity of 25 m/s.
Calculate the kinetic energy of the car.

(3)

kinetic energy = ... J

Q4.

Which row of the table is correct for both force and velocity?

(1)

	force	velocity
☐ A	scalar	scalar
☐ B	scalar	vector
☐ C	vector	scalar
☐ D	vector	vector

(Total for question = 1 mark)

Q5.

A toy car has a mass of 0.10 kg.
The toy car accelerates at 2.0 m/s^2.

Calculate the force producing this acceleration.
State the unit.

Use the equation

$F = m \times a$

(3)

force = ... unit = ...

(Total for question = 3 marks)

Q6.

Figure 8 shows a trolley at the top of a slope.

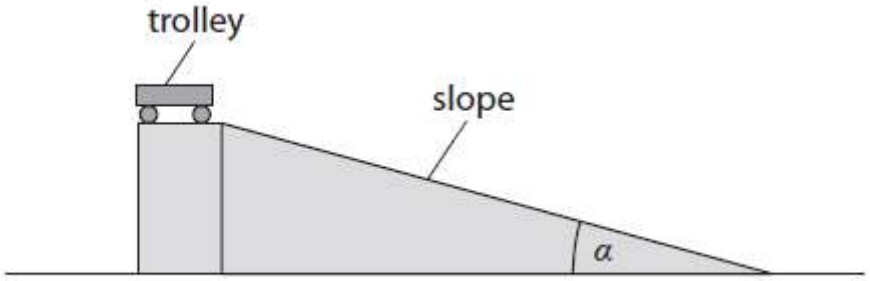

Figure 8

A student gently pushes the trolley until it just starts to roll down the slope.
The student measures the time it takes for the trolley to roll down the slope.
The student repeats this for different values of the angle *a*.
Figure 9 is a graph of the student's results.

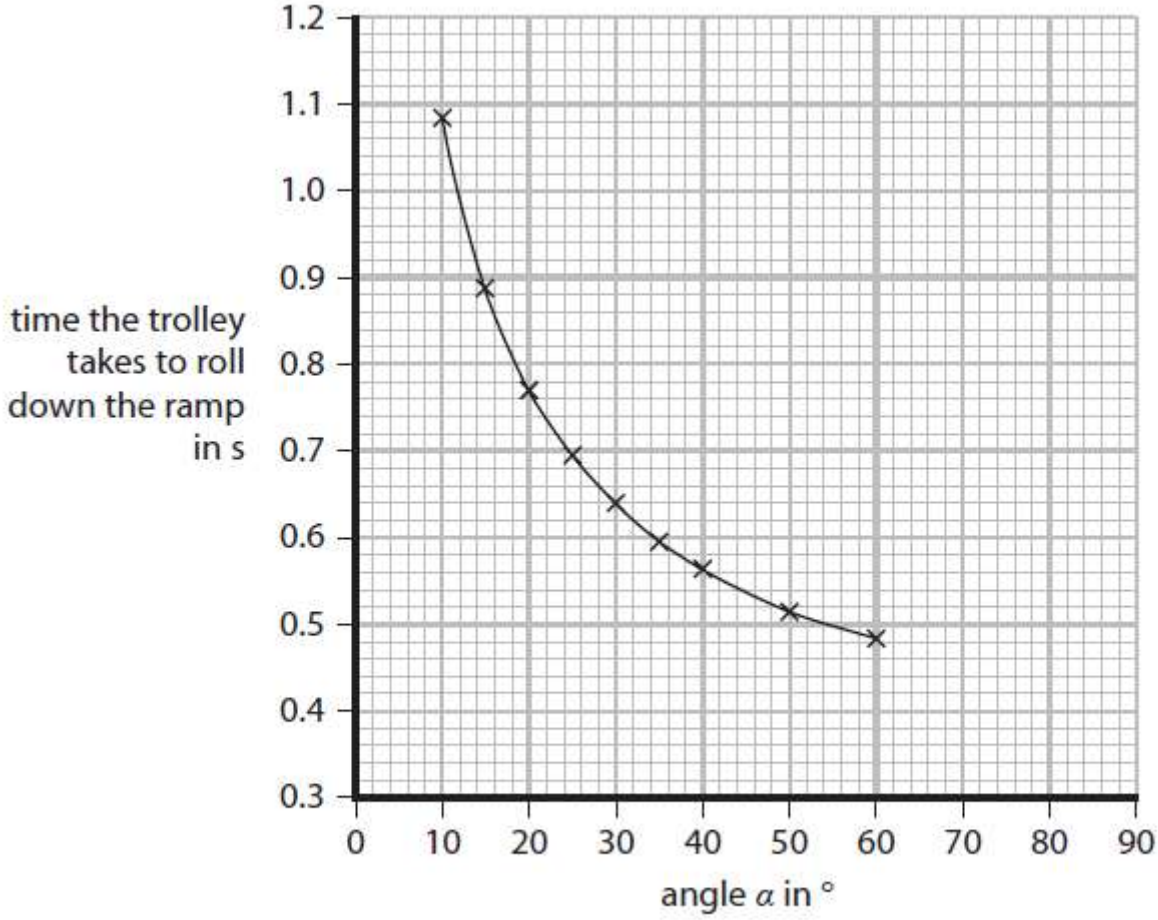

Figure 9

(i) Use the graph in Figure 9 to find the time the trolley takes to roll down the ramp when the angle *a* = 45°.

(1)

time = ... s

(ii) Use the graph in Figure 9 to estimate the time the trolley takes to roll down the ramp when the angle *a* = 80°.
Show your working on the graph.

(2)

time = ... s

(iii) The student had a choice of how to measure the time the trolley takes to roll down the ramp.
1. Use a hand-held stopwatch.
2. Use light gates at the top and bottom of the slope.

The student chose to use the light gates.
Explain why this was the correct choice.
You should refer to the data on the time axis of Figure 9 in your answer.

(2)

.....

.....

.....

.....

(Total for question = 5 marks)

Q7.

A car approaches traffic lights.
The traffic lights turn to red so the car has to stop.

Which of the following factors affects the thinking distance when the car has to stop?

(1)

☐ **A** condition of the road

☐ **B** mass of the car

☐ **C** reaction time

☐ **D** worn brakes

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A car is travelling at 10 m/s.

The driver sees a danger and stops the car.

(i) The stopping distance for the car would be smaller if the car

(1)

☐ **A** had more passengers
☐ **B** had worn tyres
☐ **C** needed new brakes
☐ **D** was travelling more slowly

Figure 4 shows a speed-time graph for the driver stopping the car.

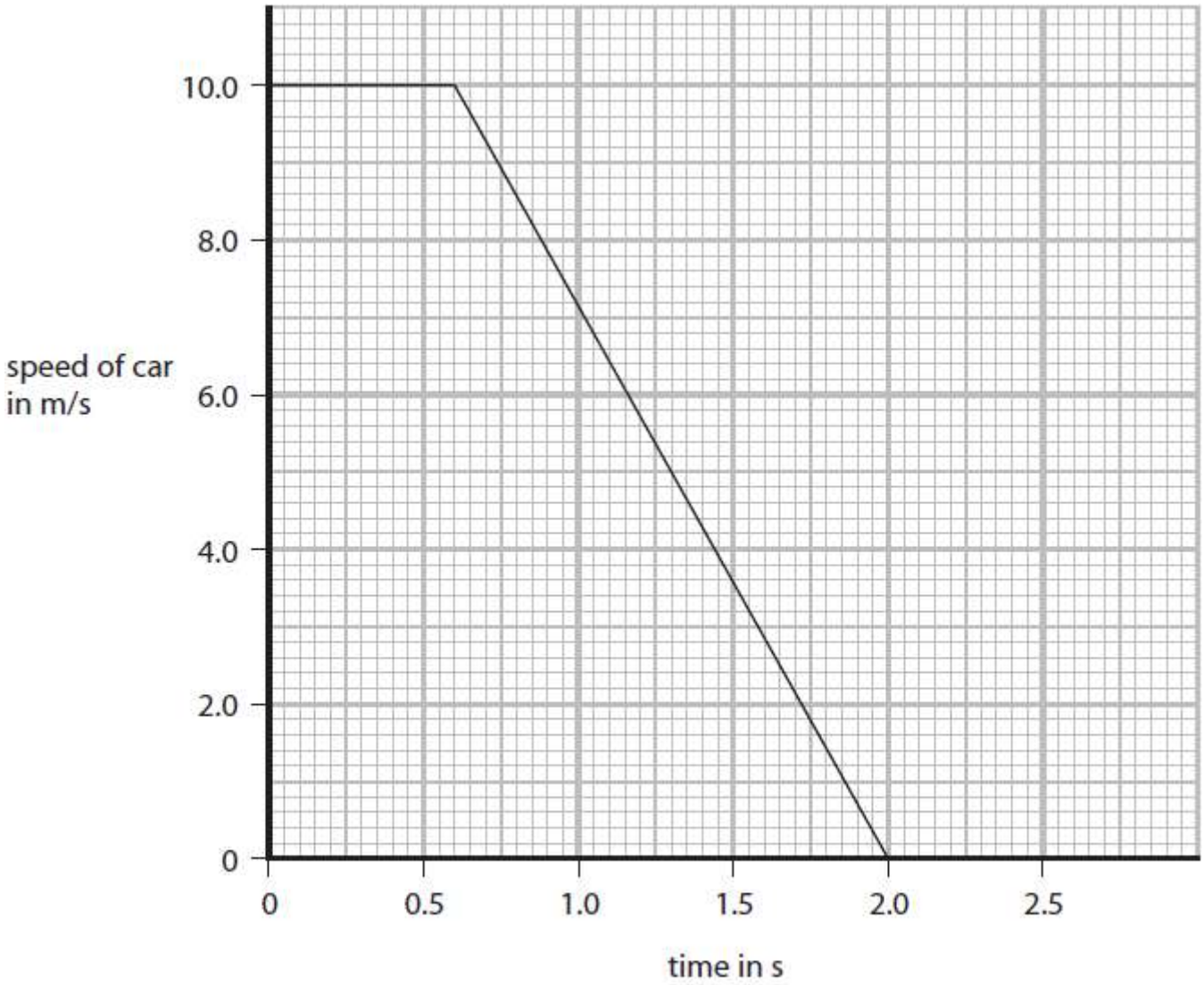

Figure 4

(ii) Use the graph to find the driver's reaction time.

(2)

reaction time = ... s

(Total for question = 3 marks)

Q9.

The stopping distance of a car is the thinking distance plus the braking distance.

A car has a device that can detect an obstacle in the road ahead.

The device is linked to a computer that can apply the brakes.

It is claimed that, in an emergency, the computer-controlled car will have a shorter stopping distance than

if the car is controlled by a human driver.

Explain why this claim could be true.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q10.

Figure 15 is a speed limit sign from a European motorway. The speeds shown are in km/h (kilometres per hour).

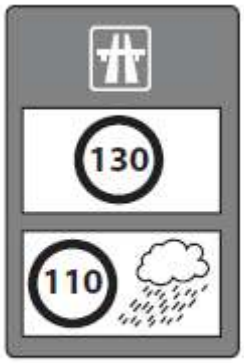

Figure 15

(i) The sign tells drivers to drive at a slower speed in wet weather.
Explain why it is safer for drivers to drive at a slower speed in wet weather.

(2)

.....

.....

.....

.....

(ii) Show that a speed of 31 m/s is less than a speed of 130 km/h.

(2)

(iii)

The driver's reaction time is the time between the driver seeing an emergency and starting to brake.

A car is travelling at a speed of 31 m/s.
The car travels 46 m between the driver seeing an emergency and starting to brake.
Calculate the driver's reaction time.
Give your answer to 2 significant figures.

(3)

driver's reaction time ..... s

(Total for question = 7 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	0.52		(1) AO3

Question number	Answer	Additional guidance	Mark
(ii)	addition and division (1) $\frac{0.35 + 0.32 + 0.38 + 0.33}{4}$ evaluation (1) 0.35 (m)	$\frac{0.35 + 0.32 + 0.52 + 0.38 + 0.33}{5}$ accept 0.345 (m) award full marks for correct answer without working. accept 0.38 for 2 marks (five results included in average	(2) AO2

Question number	Answer	Additional guidance	Mark
(iii)	Any **one** from make the slope steeper(1) add more books/blocks (1) push/pull the trolley (1)	 accept 'higher slope/high slope accept means of reducing friction e.g. use lubricant	(1) AO1

Q2.

Question number	Answer	Additional guidance	Mark
	B force A is incorrect, mass is a scalar quantity C is incorrect, energy is a scalar quantity D is incorrect, distance is a scalar quantity		**(1)** **AO1**

Q3.

Question Number	Answer	Acceptable answers	Mark
(a)	**D** driving for a long time without taking a break		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)(i)	substitution 1200 x 8(.0) (1) evaluation 9600 (J) OR 9.6 x 10^3 (J) (1)	Give full marks for correct answer with no working. 9.6 x any other power of 10 = 1 mark	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)(ii)	substitution 0.5 x 1400 x 25^2 (1) evaluation of v squared 0.5 x 1400 x 625 (1) evaluation 4.4 x 10^5 (J) (1) OR 440 000	Give full marks for correct answer with no working. accept 625 seen anywhere for this mark e.g. 875 000 gets 1 mark (forgot ½) 437 500 (J) 4.4 x any other power of 10 = 2 marks	**(3)**

Q4.

Question Number	Answer	Mark
	D | **vector** | **vector** **The only correct answer is D** ***A*** *'scalar scalar' is incorrect, both force and velocity are vectors* ***B*** *'scalar vector' is incorrect, with force being described incorrectly as a scalar* ***C*** *'vector scalar' is incorrect, with velocity being described incorrectly as a scalar*	**(1)** AO 1 1

Q5.

Question Number	Answer	Additional guidance	Mark
	substitution (1) (F =) 0.10 x 2.0	100 x 2 (using 0.10kg = 100g) reject 0.10 x 2.0^2 and the follow up evaluation (equation given should be used)	**(3)** AO 2 1
	evaluation (1) 0.2(0)	correct answer without working gets 2 marks allow 1 mark total for 2 with any other power of ten, so that includes 200 for example	
	unit (1) N	separate unit mark newtons / Newtons accept lowercase 'n' for the abbreviated unit accept kg ms^{-2} accept 200 g ms^{-2} for 3 marks	

Q6.

	Answer	Additional guidance	Mark
(i)	0.54 (s)	allow any value from 0.53 and 0.55 inclusive	(1) AO3

	Answer	Additional guidance	Mark
(ii)	curve extended to a = 80° (1) 0.45 (s) (1)	judge generously allow range 0.42 to 0.48 award full marks for the correct answer without working.	(2) AO3

	Answer	Additional guidance	Mark
(iii)	mention/idea of reaction time (1) (reaction time) about the same as the times on the graph (1)	human reaction time is about 0.2 seconds (compared with) 0.4 seconds on the graph ignore accuracy ignore "human error"	(2) AO3

Q7.

Question number	Answer	Mark
	C reaction time	(1)

Q8.

Question number	Answer	Additional guidance	Mark
(i)	D travelling more slowly A is incorrect, more passengers would increase the stopping distance B is incorrect, worn tyres would increase the stopping distance C is incorrect, if the car needed new brakes this would increase the stopping distance		(1) AO1

Question number	Answer	Additional guidance	Mark
(ii)	identification of horizontal line as reaction time (1) evaluation (1) 0.6 (s)	 award full marks for correct answer without working 0.7 scores 1 mark	(2) AO3

Q9.

Question number	Answer	Additional guidance	Mark
	an explanation linking any **two** from shorter reaction time (of computer-controlled car) (1) shorter thinking distance (1)	allow reverse arguments computer (reaction time) not affected by other relevant factors shorter distance (travelled) **before brakes applied**	**2** **AO3.1**

Q10.

	Answer	Additional guidance	Mark
(i)	an explanation linking **two** from: (wet road means) less / no friction (between tyres and road) (1) (wet weather means) increased stopping distance (1) (slower speed means) shorter braking / stopping distance (1) (dry weather / slower speed) reduces possibility of skidding / sliding / idea of losing control / crashing (1)	accept reverse arguments throughout accept road more slippery / less grip accept idea of reduced visibility accept braking or thinking distance in this context accept takes longer to slow down / stop ignore harder to brake	**(2)** **AO1**

	Answer	Additional guidance	Mark
(ii)	convert **either** distance or time (1) (31 m =) $\frac{31}{1000}$ (km) or 0.031 (km) **OR** (1 s =) $\frac{1}{3600}$ (h) = $\frac{1}{60\times60}$ (h) or 0.000 28 (h) evaluation (1) (31 m/s =) 110 (km/h)	(130 km =) 130 × 1000 (m) or 130 000 (m) **OR** (1 h =) 60 x 60 (s) or 3600 (s) (130 km/h =) 36(.1) (m/s) accept 111.6 or 112 (km/h) for 2 marks if no other marks awarded accept <u>1860 m/min</u> and <u>2167 m/min</u> for 1 mark each award full marks for the correct answer without working	**(2)** **AO2**

	Answer	Additional guidance	Mark
(iii)	select and substitute into distance travelled = average speed x time (1) 46 = 31 x t rearrangement and evaluation (1) (t=) 1.48(3) (s) evaluation given to 2 sf (1) (t =) 1.5 (s)	 $31 = \frac{46}{t}$ $(t =) \frac{46}{31}$ award two marks for the correct evaluation without working any answer written to 2 sf independent mark 1.5 scores 3 marks 1.4 scores 2 marks 1.50 scores 2 marks 0.67 scores 2 marks 1400 scores 2 marks 0.673(9) scores 1 mark 1426 scores 1 mark	**(3)** **AO2**